

L1.6 Trig Derivatives & the Chain Rule

§2.4–2.5 — *what you need to know*

The six trig derivatives

LEARN THESE THREE

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

GENERATE THESE THREE

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

“the derivative of a co-function is always negative”

Take the row on the left, put *co-* on every function, attach a minus sign. The co-function derivative is the co-function **of the derivative**: $\sec x \tan x \rightarrow \csc x \cot x$, negated.

RADIANS AND DEGREES

$\frac{dy}{dx}$ is an output change over an input change, so differentiating a trig function puts an **angular** measure and a **linear** one in the same ratio.

Linear and angular measures mixing forces radians. Where nothing mixes, the choice is free. Differentiating is itself the mixing — so every formula above is in radians.

The receipt: $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$ in radians but $\frac{\pi}{180}$ in degrees — which is why $\frac{d}{dx} \sin(x^\circ) = \frac{\pi}{180} \cos(x^\circ)$. Reading a graph or naming $\sin 30^\circ$ mixes nothing, so there you are free.

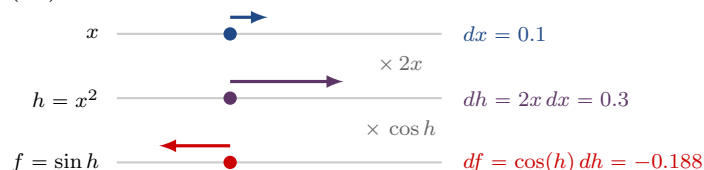
DIFFERENTIALS

dx , dy and $d(f(x))$ **live on their own**. dx is a nudge in the input, dy is the nudge that comes back out, and $\frac{dy}{dx}$ is a *ratio of two of them* — not one symbol.

$\frac{dy}{dx} = f'(x) \implies dy = f'(x) dx$. At $x = 1$ with $dx = 0.1$: $y = x + 1$ gives $dy = dx = 0.1$, the nudge straight through; $y = x^2$ gives $dy = 2x dx = 0.2$ — **the same nudge, doubled on the way out**.

THE CHAIN RULE

Nudge the input and watch the nudge come out the far end **changed by a factor at each stage**. For $y = \sin(x^2)$ at $x = 1.5$ with $dx = 0.1$:



Substitute the middle line into the bottom one — $df = \cos(x^2) \cdot 2x dx$ — and divide by dx :

$$\frac{df}{dx} = f'(h(x)) \cdot h'(x)$$

f' is **evaluated at the inside**, times the derivative of the inside. A **negative factor** sends the nudge back **out the other way**, as above.

WHICH RULE RUNS FIRST

The **outermost operation** names the first rule. Read what the expression *is* before differentiating any piece of it.

$(2x + 1)^2 x^3$	a product	product , then chain on $(2x + 1)^2$
$\frac{\sin x}{x^2 + 1}$	a quotient	quotient
$\sin(x^2 + 1)$	sine of something	cos(inside) , then chain
$(x^2 \sin x)^4$	a fourth power	power , then chain, then product inside

The **chain rule is not a competitor to the other rules** — it is what comes next. The outermost operation picks the rule you run first; chain is the instruction to then multiply by the derivative of the inside.

WATCH OUT

✗ $\frac{d}{dx}(\sin x^2) = \cos x^2 \rightarrow$ the inner derivative is missing — it is $\cos(x^2) \cdot 2x$

✗ $\frac{d}{dx}(\cos x) = \sin x \rightarrow$ co-functions carry the minus: $-\sin x$

✗ $\frac{d}{dx}(\sin^2 x) = 2 \sin x \rightarrow$ the outer function is **squaring**: $2 \sin x \cos x$

✗ $\frac{dy}{dx}$ is one symbol $\rightarrow dx$ and dy are differentials — each means something on its own

✗ the chain rule multiplies by the derivative of the inside $\rightarrow$ it is f' *evaluated at the inside*, times the derivative of the inside

✗ a composite needs one rule $\rightarrow (2x + 1)^2 x^3$ needs product first, then chain

✗ $(x^2 \sin x)^4$ starts with the chain rule $\rightarrow$ it starts with the **power** rule — chain is what comes next